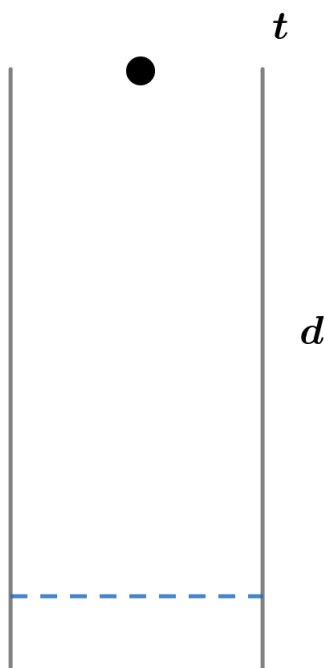


# 2019A F=ma Exam: Problem 16

Kevin S. Huang



We can divide  $t$  into two intervals, the time  $t_1$  it takes the stone to hit the bottom and the time  $t_2$  it takes for the sound to travel back up. We have

$$t = t_1 + t_2$$

$$d = \frac{1}{2}gt_1^2$$

$$d = vt_2$$

where  $v$  is the speed of sound. If we neglect the speed of sound then

$$d' = \frac{1}{2}gt^2$$

We require

$$\frac{d'}{d} = \frac{t^2}{t_1^2} < 1 + \Delta$$

so

$$\frac{t}{t_1} = \frac{t_1 + t_2}{t_1} = 1 + \frac{t_2}{t_1} < \sqrt{1 + \Delta}$$

$$\frac{t_2}{t_1} = \frac{d}{v} \sqrt{\frac{g}{2d}} < \sqrt{1 + \Delta} - 1$$

$$\frac{gd}{2v^2} < (\sqrt{1 + \Delta} - 1)^2$$

Thus,

$$d < \frac{2v^2}{g} (\sqrt{1 + \Delta} - 1)^2 = \frac{2(330 \text{ m/s})^2}{10 \text{ m/s}^2} (\sqrt{1 + 0.05} - 1)^2 = 13.3 \text{ m}$$

so the answer is C.